

Extension of Charges on Directed Systems of Boolean Algebras

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Abstract

Stone's representation theorem identifies a Boolean algebra $\mathcal{E}$ with the clopen algebra of a compact totally-disconnected space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ with the Baire σ -algebra of $S(\mathcal{E})$. Under this correspondence, the classical question — when does a charge on $\mathcal{E}$ extend to a σ -additive measure on $\sigma(\mathcal{E})$? — becomes: when does a finitely-additive measure on the clopens of $S(\mathcal{E})$ extend to a Borel measure? The answer for a single algebra is well known: continuity at $\emptyset$, equivalently tightness on the Stone space.

This paper asks the question for a *directed system* of Boolean algebras: a preordered family $\{(\mathcal{E}_i, \ell_i)\}_{i \in \mathcal{I}}$, each connected by surjective refinement maps with compatible charges. The Stone spaces $\{S(\mathcal{E}_i)\}$ form a projective system, and the extension question becomes whether the charges propagate coherently to σ -additive measures throughout.

We show three things. First, the gap $\sigma(\mathcal{E}) \setminus \mathcal{E}$ is not a defect: a level with no gap admits no nontrivial refinement above it, and $\sigma(\mathcal{E}) = \mathcal{E}$ implies algebraic triviality of the refinement order. Second, index-layer coherence conditions alone — sequential upper-directedness and compatibility — do not force σ -additivity; this confirms, in the system setting, that no finitary structural condition suffices, consistent with the classical single-algebra result. The counterexample is a compatible family on a sequentially upper-directed system in which every level permanently assigns unit mass to a sequence of events the system as a whole sees as empty. Third, we identify the correct characterisation: a family of finitely-additive charges extends to σ -additive measures at every level if and only if it is *collectively exhaustive* — a valuation-layer condition requiring that every vanishing sequence is witnessed somewhere in the refinement order. Collective exhaustion is the algebraic substitute for the tightness condition on the Stone space, requiring no topology on the outcome spaces.

1 Introduction

A Boolean algebra $\mathcal{E}$ of events, together with a finitely-additive charge $\ell : \mathcal{E} \rightarrow [0, 1]$, constitutes a Boolean charge space. Under Stone's representation theorem [5], $\mathcal{E}$ is

isomorphic to the clopen algebra of a compact totally-disconnected space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ is the Baire σ -algebra of $S(\mathcal{E})$. The extension question — when does ℓ extend to a σ -additive measure on $\sigma(\mathcal{E})$? — becomes: when is the charge on the clopens tight enough to extend to the full Baire σ -algebra? The extension criterion (Theorem 3.5) gives the answer: continuity at $\emptyset$, in the sense of Daniell–Stone [1, 6], is necessary and sufficient. No finitary condition on $\mathcal{E}$ alone can supply this: the Stone space of a non-compact Boolean algebra carries charges that do not extend, regardless of any algebraic constraint [2]. The question is whether something beyond the single algebra can.

This paper studies the extension question for a *directed system* of Boolean algebras: a preordered family $\{(\mathcal{E}_i, \ell_i)\}_{i \in \mathcal{I}}$ connected by surjective refinement maps with compatible charges. The Stone spaces $\{S(\mathcal{E}_i)\}$ form a projective system. The question becomes whether the charges propagate coherently to σ -additive measures throughout — and if so, what condition on the family is necessary and sufficient.

The paper establishes three results. The negative result (Section 3) shows that the extension question has no finitary answer even in the system setting: no chain of finite refinements resolves the obstruction. The incompleteness result (Section 4) shows that nontrivial refinement is an irreducibly finer operation: a Boolean algebra $\mathcal{E}$ admitting a nontrivial refinement contains events not pullable from below, and a level at which $\mathcal{E} = \sigma(\mathcal{E})$ admits no nontrivial refinement at all. The positive result (Section 5) identifies the correct characterisation. The natural question — whether index-layer conditions (directedness, compatibility) force σ -additivity — turns out to be ill-posed: it conflates conditions on the directed system’s structure with conditions on the charges it carries. We show that index-layer coherence does not force σ -additivity (the finite-cofinite counterexample), and that σ -additivity is equivalent to collective exhaustion, a purely valuation-layer condition (Theorem 5.3).

Section 6 locates the results and states what remains open.

1.1 Organisation

Section 2 sets up the single-algebra case: a Boolean algebra $\mathcal{E}$ with a finitely-additive charge ℓ on it, with no topology and no σ -algebra assumed.

Section 3 establishes that finitary conditions do not force σ -additivity: exhaustiveness is automatic, the genuine obstruction is continuity at $\emptyset$ for sequences escaping $\mathcal{E}$, and no finitary condition on any single level can supply this.

Section 4 establishes the incompleteness result: a nontrivial refinement introduces events not pullable from the coarser level, and a level with $\mathcal{E} = \sigma(\mathcal{E})$ admits no nontrivial refinement.

Section 5 completes the analysis: it proves the independence of index- and valuation-layer conditions and establishes the main equivalence (collective exhaustion $\iff$ σ -additive extensibility).

Section 6 locates the results within the Stone duality picture and states what remains open.

2 Single-algebra setup

We begin with a single Boolean algebra and its charge, setting aside the directed system structure to isolate the extension question.

Definition 2.1 (Boolean charge space). A *Boolean charge space* is a pair $(\mathcal{E}, \ell)$ where:

- (i) $\mathcal{E}$ is a Boolean algebra of subsets of some set O — closed under finite union, intersection, and complementation, containing $\emptyset$ and O .
- (ii) $\ell : \mathcal{E} \rightarrow [0, 1]$ is a normalized finitely-additive valuation: $\ell(\emptyset) = 0$, $\ell(O) = 1$, and $\ell(A \cup B) = \ell(A) + \ell(B)$ whenever $A \cap B = \emptyset$.

Via Stone's representation theorem, $\mathcal{E}$ is isomorphic to the clopen algebra of a compact totally-disconnected space $S(\mathcal{E})$, the Stone space of $\mathcal{E}$, and $\sigma(\mathcal{E})$ is the Baire σ -algebra of $S(\mathcal{E})$.

No σ -algebra is assumed on O . No topology is assumed.

Definition 2.2 (The gap). For a discriminability structure $(\mathcal{E}, \ell)$, the *gap* is the set $\sigma(\mathcal{E}) \setminus \mathcal{E}$ — events in the σ -algebra generated by $\mathcal{E}$ that are not themselves in $\mathcal{E}$.

The gap consists of events reachable by countable Boolean operations on $\mathcal{E}$ but not contained in $\mathcal{E}$ itself.

Definition 2.3 (Extension). We say ℓ *extends* to $\sigma(\mathcal{E})$ if there exists a σ -additive measure μ on $\sigma(\mathcal{E})$ with $\mu|_{\mathcal{E}} = \ell$. The extension is *unique* if any two such μ agree on all of $\sigma(\mathcal{E})$.

The central question of this paper is: what properties of $(\mathcal{E}, \ell)$ are necessary and sufficient for such an extension to exist and be unique?

3 Finitary conditions do not force σ -additivity

3.1 Exhaustiveness is automatic

The Stone space $S(\mathcal{E})$ of a Boolean algebra $\mathcal{E}$ is compact by construction. But compactness of $S(\mathcal{E})$ does not force a given charge on $\mathcal{E}$ to extend to a Borel measure on $S(\mathcal{E})$: compactness is a property of the space, not the measure. The measure-theoretic condition is tightness — equivalently, continuity at $\emptyset$. The question is whether this can be witnessed finitely. It cannot.

The first natural candidate is exhaustiveness.

Definition 3.1 (Exhaustiveness). A finitely-additive valuation $\ell : \mathcal{E} \rightarrow [0, 1]$ is *exhaustive* if $\ell(E_n) \rightarrow 0$ for every disjoint sequence $(E_n)_{n \geq 1} \subset \mathcal{E}$.

Proposition 3.2. *Every normalized finitely-additive valuation on a Boolean algebra is exhaustive.*

Proof. Let (E_n) be a disjoint sequence in $\mathcal{E}$. For any N , $\sum_{n=1}^N \ell(E_n) = \ell(\bigcup_{n=1}^N E_n) \leq \ell(O) = 1$ by finite additivity and normalization. Hence $\sum_{n=1}^\infty \ell(E_n) \leq 1$, so $\ell(E_n) \rightarrow 0$. $\square$

Exhaustiveness carries no information about extensibility. It is a consequence of normalization alone, independent of any coherence condition or structural hypothesis.

3.2 Continuity at $\emptyset$

The correct extension condition is continuity from above at $\emptyset$.

Definition 3.3 (Continuity at $\emptyset$). A finitely-additive valuation $\ell : \mathcal{E} \rightarrow [0, 1]$ is *continuous at $\emptyset$* if: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \dots$ in $\mathcal{E}$ with $\bigcap_{n=1}^\infty E_n = \emptyset$ in $\sigma(\mathcal{E})$, we have $\ell(E_n) \rightarrow 0$.

Remark 3.4. The intersection $\bigcap_n E_n$ in Definition 3.3 is taken in $\sigma(\mathcal{E})$, not in $\mathcal{E}$. The condition is therefore not verifiable from within $\mathcal{E}$ alone: confirming $\bigcap_n E_n = \emptyset$ requires access to $\sigma(\mathcal{E})$. This is precisely what the regress below captures.

Theorem 3.5 (Extension criterion). *A normalized finitely-additive $\ell : \mathcal{E} \rightarrow [0, 1]$ extends to a σ -additive measure on $\sigma(\mathcal{E})$ if and only if ℓ is continuous at $\emptyset$. The extension is then unique.*

Proof. Standard: see Halmos [4] or Fremlin [3] Vol. 1. The forward direction is immediate from σ -additivity of the extension. For the converse, continuity at $\emptyset$ is precisely the condition used in the standard proof of the Carathéodory extension to pass from finite additivity to σ -additivity on the generated σ -algebra. $\square$

3.3 The regress

In a directed system, one might hope that a finer level $j \geq i$ can witness the emptiness that level i cannot see, and that compatibility then forces continuity at $\emptyset$ at level i .

Proposition 3.6 (The regress). *Let $j \geq i$ in a directed system with refinement map $\pi : O_j \rightarrow O_i$. Suppose $E_1 \supseteq E_2 \supseteq \dots$ in $\mathcal{E}_i$ with $\bigcap_n \pi^{-1}(E_n) = \emptyset$ in $\mathcal{E}_j$. Compatibility gives $\ell_i(E_n) = \ell_j(\pi^{-1}(E_n))$. But this does not force $\ell_j(\pi^{-1}(E_n)) \rightarrow 0$: the same obstruction reappears at level j , since finite additivity on $\mathcal{E}_j$ alone does not imply continuity at $\emptyset$ for sequences decreasing to an element of $\sigma(\mathcal{E}_j)$.*

The regress is structural. Pushing the witnessing one level up does not resolve it: it only relocates it. No finite chain of refinements can supply continuity at $\emptyset$ from purely finitary conditions.

Corollary 3.7. *No condition stateable purely in terms of the Boolean algebra structure of $\mathcal{E}_i$ and the finitely-additive charges $\{\ell_j\}_{j \geq i}$ at finitely many levels is necessary and sufficient for ℓ_i to extend to $\sigma(\mathcal{E}_i)$.*

Something external to any single level is required. The three candidates are: topology (compactness witnesses emptiness), a colimit algebra (a direct limit in the directed system whose event algebra contains the limit events), or an explicit continuity axiom (honest but circular). The remainder of this paper develops the colimit option, which is intrinsic to the directed system framework.

4 The structure of nontrivial refinement

Before turning to the positive result, we establish what nontrivial refinement actually entails at the level of the Boolean algebras.

Definition 4.1 (Nontrivial refinement). Let $\pi : O_j \rightarrow O_i$ be a surjective refinement map between levels $j \geq i$ in a directed system, inducing the Boolean algebra homomorphism $\pi^{-1} : \mathcal{E}_i \rightarrow \mathcal{E}_j$. The refinement is *nontrivial* if there exists $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$ — that is, A is not the preimage of any event at level i .

Proposition 4.2 (Nontrivial refinement introduces new events). *If $j \geq i$ is a nontrivial refinement, then $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$: the finer algebra strictly extends the pullback of the coarser.*

Proof. Immediate from the definition: nontriviality asserts the existence of $A \in \mathcal{E}_j \setminus \pi^{-1}(\mathcal{E}_i)$. $\square$

Remark 4.3. The contrapositive is equally important: if $\mathcal{E}_j = \pi^{-1}(\mathcal{E}_i)$ for every $j \geq i$, then every finer level distinguishes exactly the events pullable from level i — the refinement order above i is algebraically trivial. Nontriviality of refinement is therefore the condition that the directed system is genuinely increasing in discriminative power.

Remark 4.4 (Relation to the gap). The gap $\sigma(\mathcal{E}_i) \setminus \mathcal{E}_i$ is a separate condition: it concerns events reachable by countable operations on $\mathcal{E}_i$ that are not themselves in $\mathcal{E}_i$. When $\mathcal{E}_i$ is not already a σ -algebra — in particular when O_i is uncountable and $\mathcal{E}_i$ does not contain all subsets of O_i — this gap is nonempty. Under Stone duality, the gap corresponds to the Baire sets of $S(\mathcal{E}_i)$ that are not clopen: a level with $\mathcal{E}_i = \sigma(\mathcal{E}_i)$ would correspond to a Stone space whose entire Baire σ -algebra consists of clopens, a highly degenerate condition. In the settings of interest — outcome spaces that are not finite discrete — the gap is nonempty, and the extension question is nontrivial.

5 Collective exhaustion and the characterisation of σ -additivity

The negative result established that no finitary or index-layer condition can force σ -additivity. This section completes the picture: we identify the exact valuation-layer condition that characterises σ -additive extensibility, show it is independent of the index-layer conditions, and prove the equivalence.

Proposition 5.1 (Independence of index and valuation layers). *Sequential upper-directedness together with compatibility of finitely-additive charges does not force σ -additivity. There exists a compatible family of normalized finitely-additive charges on a sequentially upper-directed directed system that fails σ -additive extension at every level.*

Proof. Let $\iota = \mathbb{N} \cup \{\omega\}$ with $n \leq \omega$ for all $n \in \mathbb{N}$, directed by this order. This is sequentially upper-directed: ω is a universal upper bound for every sequence. Take all outcome spaces to be $\mathbb{Q}$, all event algebras to be the finite-cofinite algebra $\mathcal{E} = \{E \subseteq \mathbb{Q} : E \text{ is finite or } E^c \text{ is finite}\}$, and all refinement maps to be the identity. Define $\ell(E) = 0$ if E is finite and $\ell(E) = 1$ if E is cofinite. This is a normalized finitely-additive content, and compatibility holds trivially.

Fix an enumeration $q : \mathbb{N} \rightarrow \mathbb{Q}$. Let $F_n = \{q(0), \dots, q(n)\}$. Each F_n is finite, so $\ell(F_n) = 0$ for all n . Since q is surjective, $\bigcup_n F_n = \mathbb{Q}$, and $\ell(\mathbb{Q}) = 1$. σ -subadditivity would give $1 \leq \sum_n \ell(F_n) = 0$: contradiction. So ℓ fails σ -additive extension at every level. $\square$

The counterexample establishes that index-layer coherence is insufficient: the σ -algebra is not derivable from structural conditions on the directed system alone. The charge is perfectly coherent within the finite-cofinite algebra; the failure is a valuation-layer pathology invisible to the index structure.

Definition 5.2 (Collectively exhaustive). A compatible family $\{\ell_i\}$ of normalized finitely-additive charges on a directed system is *collectively exhaustive* if: for every index $i \in \mathcal{I}$ and every decreasing sequence $E_1 \supseteq E_2 \supseteq \dots$ of events in $\mathcal{E}_i$ with $\bigcap_n E_n = \emptyset$, there exists $j \geq i$ such that $\ell_j(\pi_{ji}^{-1}(E_n)) \rightarrow 0$.

Collective exhaustion is a *valuation-layer* condition: it places a requirement on how the charges behave, not on the index structure of the directed system. It says that no level can assign persistent mass to events that the system as a whole sees as empty. The counterexample above fails collective exhaustion: for the sequence $E_n = \mathbb{Q} \setminus F_n$ (cofinite sets shrinking to $\emptyset$), every level assigns charge 1 to $\pi_{ji}^{-1}(E_n) = E_n$, and no witness exists anywhere in the system.

Theorem 5.3 (Collective exhaustion characterises σ -additivity). *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system indexed by $\mathcal{I}$. The following are equivalent:*

(i) *The family is collectively exhaustive.*

(ii) *ℓ_i extends uniquely to a σ -additive measure on $\sigma(\mathcal{E}_i)$ for every $i \in \mathcal{I}$.*

Proof. (i) $\Rightarrow$ (ii). Fix $i \in \mathcal{I}$ and a decreasing sequence $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By collective exhaustion, find $j \geq i$ with $\ell_j(\pi_{ji}^{-1}(E_n)) \rightarrow 0$. By compatibility, $\ell_i(E_n) = \ell_j(\pi_{ji}^{-1}(E_n)) \rightarrow 0$. So ℓ_i is continuous at $\emptyset$. By the extension criterion (Theorem 3.5), ℓ_i extends uniquely to a σ -additive measure on $\sigma(\mathcal{E}_i)$.

(ii) $\Rightarrow$ (i). Suppose ℓ_i extends to μ_i for every $i \in \mathcal{I}$. Let $E_n \searrow \emptyset$ in $\mathcal{E}_i$. Then $\ell_i(E_n) = \mu_i(E_n) \rightarrow \mu_i(\emptyset) = 0$ by σ -additivity of μ_i and normalization (which ensures $\mu_i(E_0) \leq \mu_i(O_i) = 1 < \infty$, the finiteness condition for continuity from above). Take $j = i$. $\square$

Remark 5.4 (The dissolution). Proposition 5.1 shows that index-layer conditions (sequential upper-directedness, compatibility) do not force σ -additivity. These layers are independent: the index layer (the preorder, refinement maps, sequential upper-directedness) and the valuation layer (the charges, compatibility, continuity at $\emptyset$) carry different information.

The correct question is: what valuation-layer condition characterises σ -additive extensibility? Theorem 5.3 answers it: collective exhaustion. It is not an axiom imposed from outside — it is the exact condition separating directed systems in which every convergence is witnessed somewhere in the refinement order from those in which it is not.

Sequential upper-directedness plays no role in Theorem 5.3: it is a per-level result. Sequential upper-directedness enters when one asks whether the per-level extensions assemble into a global probability space over Ω — that is the next question.

6 Discussion

Under Stone duality, $\mathcal{E}_i$ corresponds to the clopen algebra of a compact totally-disconnected space and $\sigma(\mathcal{E}_i)$ to its Baire σ -algebra; the extension question is whether a charge on the clopens extends to a Borel measure. The present paper answers this for directed systems without topological hypotheses on the outcome spaces: collective exhaustion is the necessary and sufficient condition, playing the role that tightness plays on the Stone space.

Theorem 5.3 is a per-level result. Sequential upper-directedness plays no role in it: that condition enters when asking whether the per-level extensions $\{\mu_i\}$ assemble into a single σ -additive measure on the projective limit $\Omega = \varprojlim O_i$. For compact outcome spaces this follows from the Prokhorov extension theorem; for standard Borel spaces from Musiał's theorem via perfectness. The topology-free case remains open.

Proposition 4.2 establishes that nontrivial refinement strictly extends the pullback algebra: $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$. The gap $\sigma(\mathcal{E}_i) \setminus \mathcal{E}_i$ is nonempty whenever $\mathcal{E}_i$ is not already a σ -algebra (Remark 4.4). Constructing a specific element of this gap from the refinement

data — connecting to the Stone space picture where the gap corresponds to Baire sets not clopen in $S(\mathcal{E}_i)$ — is the concrete content of the incompleteness claim.

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